01

BOUNDARY INTELLIGENCE ENGINE

Pre-race strategic risk mapping for track-limit compliance

> Motorsport as proof of concept — not the final destination.

02

A SPATIAL-COMPLIANCE PROBLEM, NOT JUST AN F1 PROBLEM

> "Motorsport is no longer the destination —
> it is the proof of concept."

THELITERAL ASK
A car must stay within a track boundary at high speed; a system must know when it's crossed.

THE UNDERLYING PRINCIPLE
Quantify the risk of a boundary violation for a moving object — ideally before it happens.

THE COMMON TRAP
The conventional reactive pipeline: detect car → detect line → flag crossing → replay.

03

WHY THE DIRECT APPROACH FALLS SHORT

FORENSIC, NOT PREVENTIVE

Reports violations only after the fact, no chance to react.

CAMERA DEPENDENCE LIMITS COVERAGE

Reliant on camera placement, angle, visibility, and latency.

THE VIOLATION WINDOW IS TINY

At racing speeds, boundary excursions cover very few frames, making precise detection difficult.

04

FROM INCIDENT REPORTS TO RACE-DAY STRATEGY

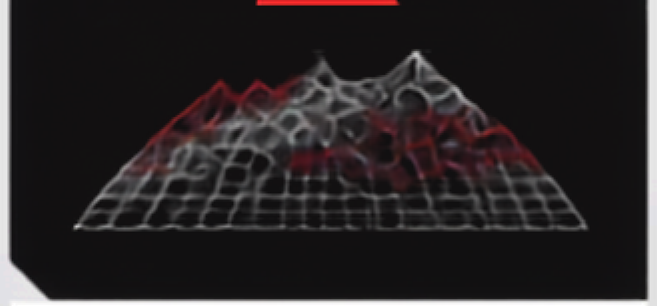
"Don't tell stewards a violation happened.
Tell race engineers, before the race,
which corners are likely to produce violations —
and what it costs in lap time to stay safely inside the limit."

01

PRACTICE SESSION RISK MAPPING

Margin-toboundary distribution per corner, per lap.

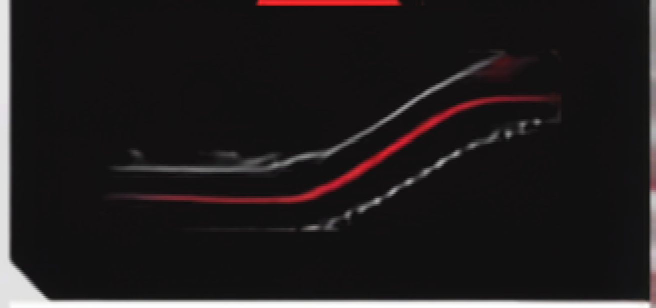
02



SIMULATION / WHAT-IF LAYER

Project risk under race conditions — tyre wear, fuel, weather.

03



TRACK-LIMIT STRATEGY

Risk-vs-lap-time trade-off, per corner, for race engineers.

05

DETECTION & TRACKING PIPELINE

PRACTICE SESSION VIDEO

↓

DETECTION (YOLO)

↓

TRACKING (BYTETRACK)

↓

GEOMETRY ENGINE
VEHICLE FOOTPRINT VS. BOUNDARY

↓

MARGIN DATA (PER LAP)

TRACK MODEL

manually calibrated track boundary per corner (MVP simplification)

STATE MACHINE

SAFE → BORDERLINE → VIOLATION → RECOVERED

OUTPUT

confidence-scored margin-toboundary distribution, per corner

06

STRATEGIC RISK & SIMULATION LAYER

01 **RISK PROFILE**
per corner

02 **RACE SCENARIOS**
tyre, fuel, weather

03 **WHAT-IF ANALYSIS**
distance-based

04 **RECOMMENDATION**
risk vs. lap time

“

Turn 9 shows
the highest predicted track-limit
risk
in the second stint.

A wider entry line
could reduce risk at
an estimated lap-time
cost of X."

07

FROM VIOLATION DETECTION TO STRATEGIC INTELLIGENCE

REACTIVE DETECTION

EVENT-LEVEL

Detect vehicle
Detect crossing
Generate replay

ENHANCED DETECTION

EVENT-LEVEL

Confidence score
Explainability

BOUNDARY INTELLIGENCE ENGINE

Margin analysis
Corner risk
Simulation
Strategic call

STRATEGY-LEVEL

08

TECH STACK

MVP-scoped, nothing over-engineered

COMPUTER VISION	Python • OpenCV • PyTorch • YOLO ByteTrack • FFmpeg
BACKEND	FastAPI
STORAGE	SQLite
FRONTEND	React / TypeScript • Vite Tailwind • shadcn/ui
SESSION PROGRESS	WebSocket (session review)

Deliberately avoids real-time public infrastructure dependencies.

WHAT SHIPS NOW VS. LATER

IN SCOPE

- Single track, practice-session footage
- Manual boundary calibration per corner
- Car detection + tracking (YOLO + ByteTrack)
- Per-corner margin-toboundary distribution
- Single-variable race-condition simulation
- Risk-vs-lap-time recommendation output

FUTURE SCOPE

- Real-time in-race detection or alerts
- Automatic boundary detection
- Multi-track / multi-season data pooling
- Multi-camera fusion
- Production-grade / edge deployment

FROM MOTORSPORT INTELLIGENCE TO A GENERAL BOUNDARY ENGINE

PHASE 2

Automatic boundary calibration multi-season pooling uncertainty-aware risk scoring

PHASE 3

Extend the same boundary-risk engine to fixed-camera environments such as ports, warehouses and industrial sites

> "Instead of telling stewards a violation happened, we tell race engineers, before the race, which corners are likely to produce violations — and what it costs to stay inside the limit."